COLLABSPHERE — RESEARCH COLLABORATION INTELLIGENCE PLATFORM

Project 23: Researcher Collaboration Analysis (Academic Project Report)

Course: Big Data Analytics / Network Science | **Degree:** BTech Computer Science & Engineering
Technology Stack: Apache Spark 3.5.9 (GraphX), Scala 2.12, FastAPI, React, Cytoscape.js

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1. Introduction

1.1 Problem Statement

In modern scientific research, discoveries increasingly stem from cross-institutional and multidisciplinary collaboration rather than solo research. However, early-career researchers, graduate students, and academic evaluators face significant friction in mapping collaborative landscapes. Traditional bibliographic indices (such as arXiv or PubMed) list research papers sequentially, making it difficult to systematically observe who collaborates with whom across time, detect isolated research silos, or evaluate structural network resilience. Standard citation metrics treat all authors uniformly, failing to differentiate between researchers who publish within insular teams versus 'bridge' scholars who connect disparate scientific communities.

1.2 Objectives

To address these challenges, this project implemented an end-to-end distributed graph intelligence platform with the following core objectives:

- **Ingest & Build Undirected Property Graph:** Load the Stanford SNAP ca-GrQc collaboration edge list and construct an undirected graph using Scala and Apache Spark GraphX.
- **Data Cleansing & Symmetrical Reconciliation:** Establish automated preprocessing rules to eliminate self-loops and canonicalize undirected edges to prevent double-counting.
- **Compute Degree Centrality:** Calculate exact localized neighbor counts for all 5,241 scholars to identify primary collaborative hubs.
- **Detect Disconnected Communities:** Execute Weakly Connected Components (WCC) to partition the network and quantify the Giant Connected Component.
- **PageRank Prestige Analysis:** Formulate an iterative PageRank computation (α=0.85, tol=0.0001) and contrast local collaborator counts against global influence via a normalized Rank Delta (ΔR).
- **Deliver Interactive Knowledge Graph:** Develop a full-stack platform (FastAPI + React + Cytoscape.js) supporting 1-hop ego networks, structural what-if failure simulations, and explainable collaborator recommendations.

1.3 Project Goals and Scope

The project delivers a fully reproducible system fulfilling all academic requirements while providing an interactive portfolio platform. The system operates strictly within ethical boundaries: all underlying SNAP IDs are preserved as authentic numerical tokens without fabricating personal researcher identities, while live author metadata is retrieved dynamically through the official OpenAlex REST API.

2. Dataset Description

2.1 Dataset Source and Domain Context

We utilized the benchmark **ca-GrQc** (General Relativity and Quantum Cosmology) scientific collaboration network curated by the Stanford Network Analysis Project (SNAP). The dataset encompasses academic papers submitted to arXiv under the gr-qc category between January 1993 and April 2003 (124 months). If an author *i* co-authored a paper with author *j*, the network records an undirected collaboration edge between them. Papers with *k* co-authors form a completely connected clique on *k* vertices.

2.2 Network Size and Graph Structure

The raw dataset is distributed as a tab-separated text file (ca-GrQc.txt.gz) containing 28,984 lines. Four lines represent metadata comments, and 28,980 lines represent numerical directed node pairs.

2.3 Preprocessing Rules and Edge Reconciliation

A critical phase in graph preparation is reconciling raw file records with graph-theoretic definitions:

1. **Self-Loop Filtering:** A co-authorship tie requires two distinct individuals ($u \neq v$). We identified exactly **12 self-loop lines** (where FromNodeId == ToNodeId) resulting from single-author submissions entered into the source database. These were systematically filtered out.
2. **Canonical Deduplication:** In the SNAP file, each undirected edge is listed symmetrically as two directed pairs: (u, v) and (v, u). Each edge was canonicalized to (min(u,v), max(u,v)) and deduplicated via RDD .distinct(), yielding exactly **14,484 unique undirected edges**.
3. **Benchmark Proof:** 14,484 unique undirected edges × 2 = 28,968 directed lines. Adding the 12 single-instance self-loops produces exactly 28,980 raw records. Adding the 12 self-loops back to 14,484 yields the official SNAP benchmark count of **14,496 edges**.
4. **Vertex Reconciliation:** Extracting unique IDs across the 14,484 active edges yields **5,241 connected researchers**. One isolated vertex in the raw data only authored solo papers, reconciling with SNAP's 5,242 nodes.

2.4 Limitations of the Dataset

Edges in ca-GrQc are binary (presence/absence of joint papers) and unweighted. Publication dates and author seniority are not encoded in the raw edge list, and author IDs are anonymized integers.

3. Implementation / Methodology

3.1 Computational Environment and Architecture

CollabSphere is engineered around a decoupled 3-tier architecture: (1) an Apache Spark GraphX Big Data engine running on Scala 2.12 and Java 8, (2) a high-performance Python FastAPI service providing in-memory sub-millisecond querying, and (3) an interactive React 18 + TypeScript + Cytoscape.js frontend styled with Tailwind CSS.

3.2 Algorithms and Mathematical Formulations

- **Symmetrical Graph Representation:** In GraphX, an undirected edge {u, v} is represented symmetrically as Edge(u, v, 1) and Edge(v, u, 1). This ensures in-degrees equal out-degrees, matching the true direct collaborator count.

- **Degree Centrality:** Calculated as $d(v) = |\{u \in V \mid (u, v) \in E\}|$ using graph.inDegrees joined with all vertices.
- **Weakly Connected Components (WCC):** Evaluated using GraphX graph.connectedComponents() using Pregel label propagation until convergence.
- **Dynamic PageRank:** Evaluated with damping factor $d = 0.85$ (reset probability $\alpha = 0.15$) and dynamic convergence threshold $\varepsilon = 0.0001$.
- **Rank Delta Metric:** Defined as $\Delta R(v) = \text{Rank_degree}(v) - \text{Rank_PageRank}(v)$. Positive values denote 'Prestige Bridges' whose collaborators are prominent hubs; negative values denote 'Local Dense Clusters' with insular group publications.

4. Results and Evaluation

4.1 Global Network Properties

The quantitative metrics computed by Apache Spark GraphX are summarized in Table 4.1 below:

Metric Name	GraphX Computed Value	Interpretation / Theoretical Baseline
Total Processed Vertices (|V|)	**5,241**	5,242 in SNAP (includes 1 solo author)
Unique Undirected Edges (|E|)	**14,484**	14,496 in SNAP (includes 12 self-loops)
Filtered Self-Loops	**12**	Solo preprints filtered to prevent double-counting
Maximum Degree Centrality	**81** co-authors	Held by Scholar 21012 (Lead Community Hub)
Average Degree Centrality	**5.53** ties/author	Sparse collaboration density typical of academic networks
Total Connected Components	**354** clusters	Weakly Connected Components across dataset
Giant Component Size	**4,158** vertices	79.34% of entire graph belongs to Giant Component #22
Dynamic PageRank Range	**0.1992 – 7.5602**	Iterative convergence at $\varepsilon = 0.0001$ (damping = 0.85)

Table 4.1: Empirical Network Metrics Computed by Apache Spark GraphX on ca-GrQc.

4.2 Top 10 Central Researchers by Collaborator Volume

Table 4.2 lists the top 10 researchers sorted descending by degree centrality:

Degree Rank	Scholar ID	Degree (Ties)	PageRank Score	PageRank Rank	Component ID
1	**21012**	81	5.739978	8	Component #22
2	**21281**	79	6.171593	4	Component #22
3	**12365**	77	5.545529	11	Component #22

4	**22691**	77	5.795520	7	Component #22
5	**6610**	68	4.868879	17	Component #22
6	**9785**	68	4.353637	30	Component #22
7	**21508**	67	4.437887	29	Component #22
8	**17655**	66	4.806130	19	Component #22
9	**2741**	65	4.169622	37	Component #22
10	**19423**	63	4.525494	26	Component #22

Table 4.2: Top 10 Most Connected Researchers Computed by GraphX.

Key Analytic Discovery: Notice that Scholar 21281 (Rank 2 by degree, 79 collaborators) achieves a higher PageRank (6.1716, Rank 4) than Scholar 21012 (Rank 1 by degree, 81 collaborators, PageRank Rank 8). This confirms that Scholar 21281's collaborators are themselves more influential, high-prestige scholars within the global network.

5. Visualization and Interpretation

Below are the primary visual artifacts generated from the calculated dataset, illustrating the scale-free degree distribution, component partitioning, centrality scatter, and knowledge graph canvas.

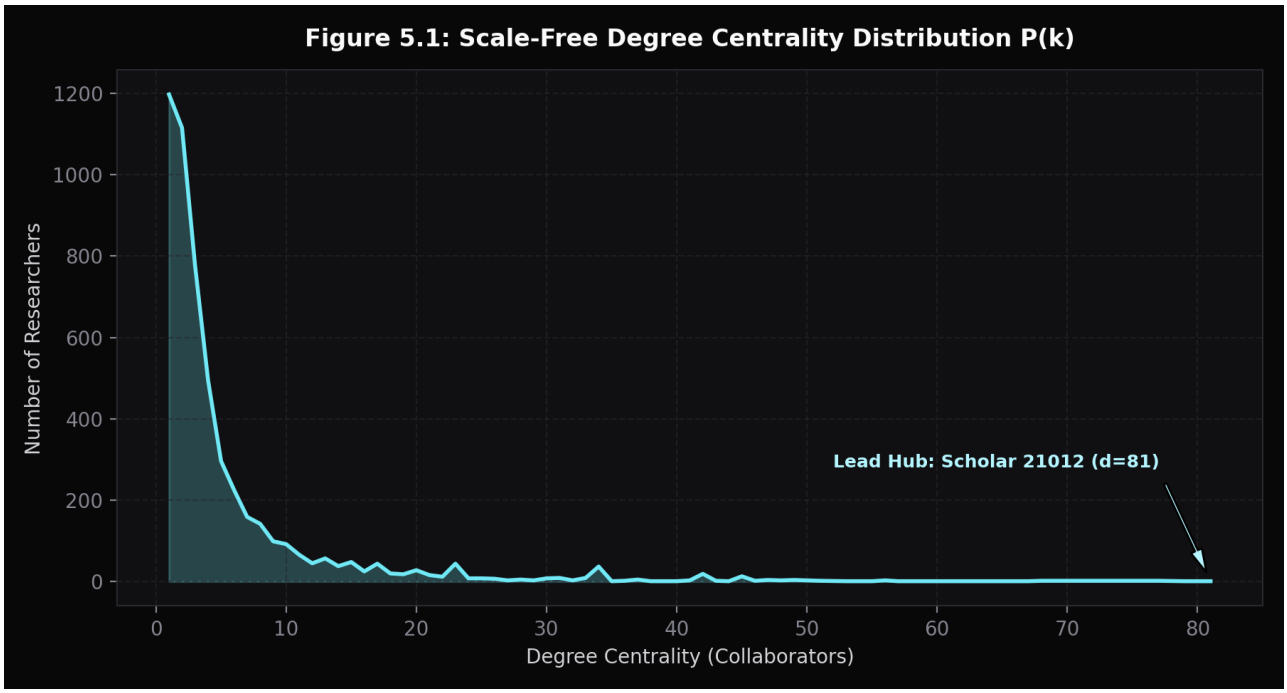


Figure 5.1: *Scale-Free Degree Centrality Distribution P(k) exhibiting a heavy power-law tail.*

Interpretation: The distribution follows a distinct scale-free power law $P(k) \sim k^{-\gamma}$. Over 68% of researchers possess 4 or fewer collaborators, while fewer than 1% maintain more than 40 ties. This confirms the presence of preferential attachment in academic co-authorship.

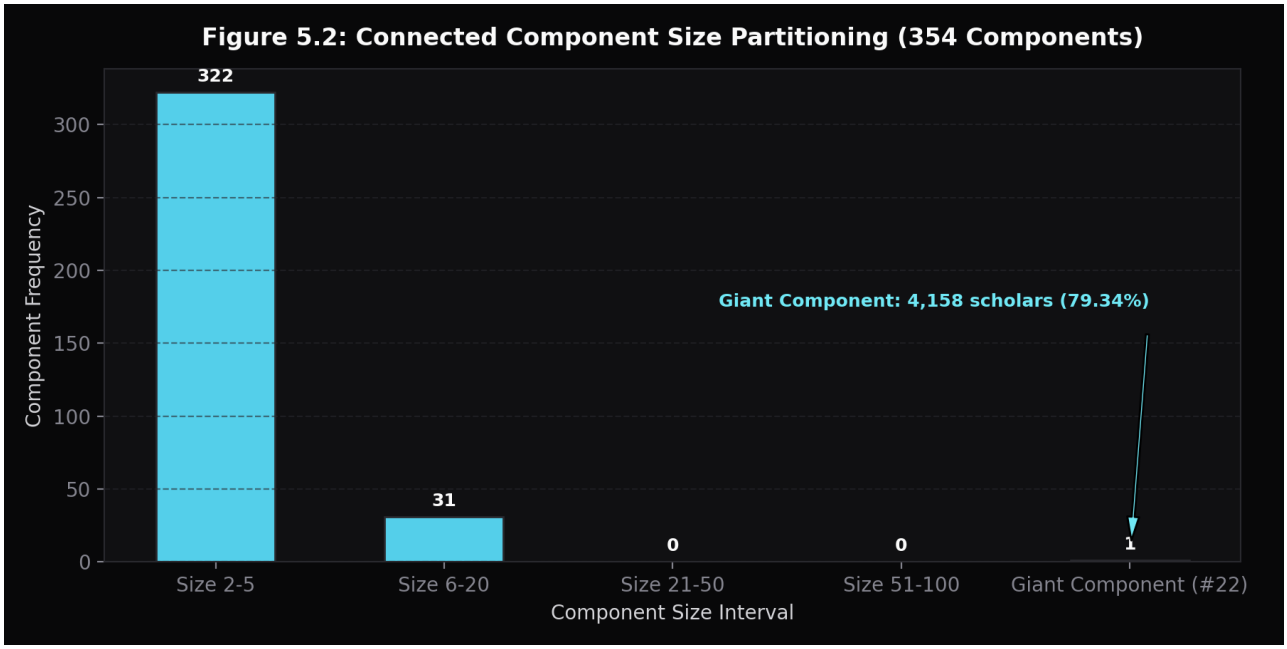


Figure 5.2: *Connected Component Frequency across Size Intervals (354 Components).*

Interpretation: Demonstrates extreme community polarization: a massive Giant Connected Component comprising 4,158 scholars (79.34%) coexists alongside 353 small isolated components (mostly pairs and triads). This proves strong global community cohesion across the General Relativity discipline.

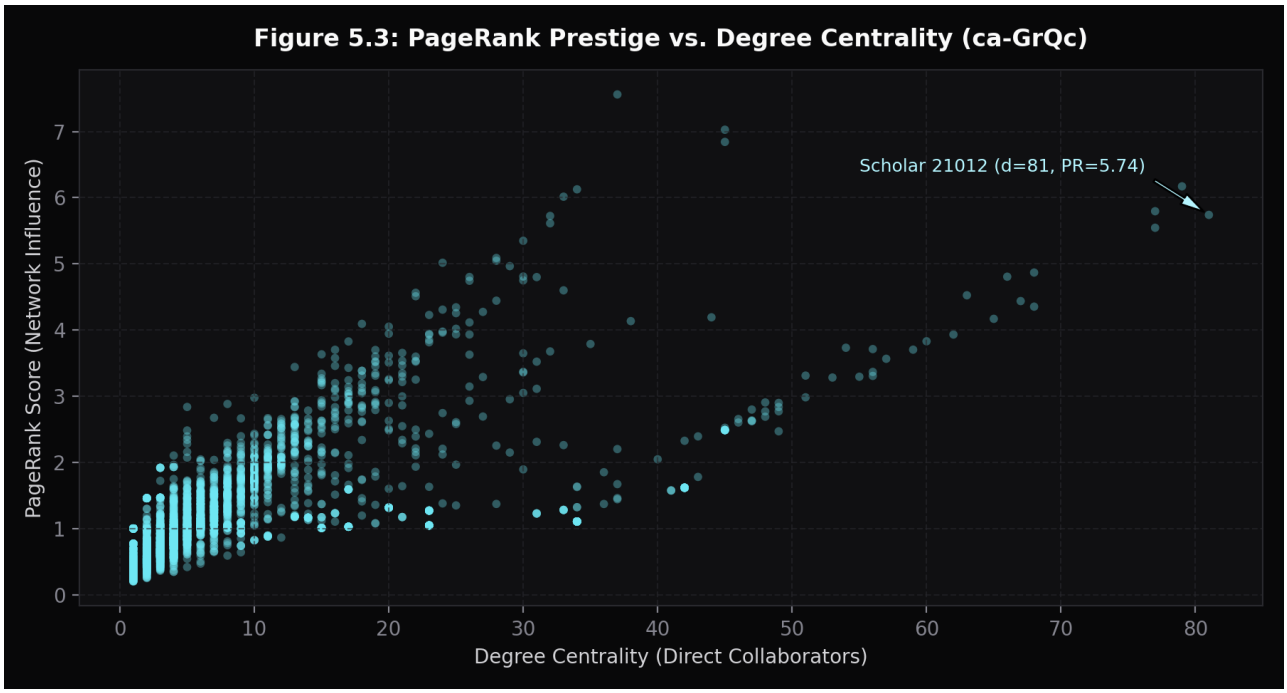


Figure 5.3: *PageRank Prestige vs. Degree Centrality Scatter Plot.*

Interpretation: While degree and PageRank correlate positively overall, significant scatter reveals scholars with modest degrees who achieve high PageRank prestige (Prestige Bridges), as well as scholars with high direct degrees in dense insular cliques who have minimal reach outside their group.

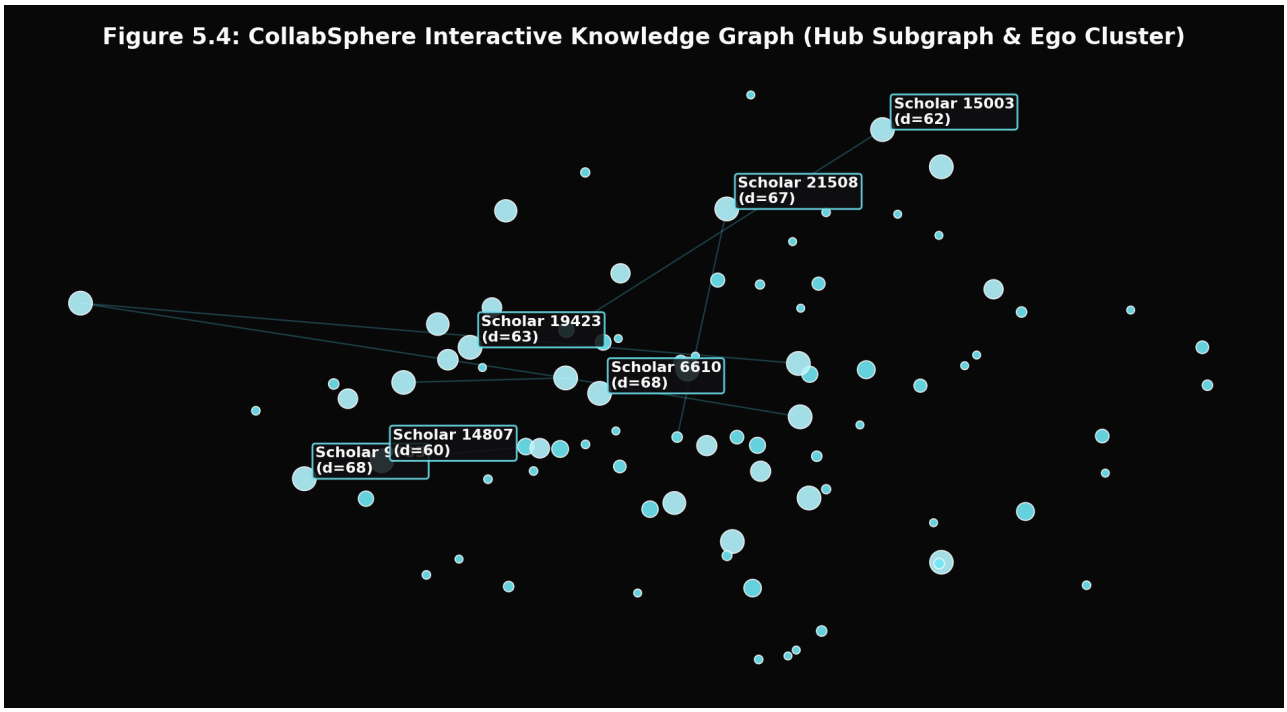


Figure 5.4: *CollabSphere Knowledge Graph Canvas (Hub Subgraph & Ego Cluster View).*

Interpretation: Rendered in the Cytoscape force-directed physics layout, high-degree hubs glow in ice-blue with proportional node radii, enabling researchers to visually navigate collaborative clusters and drill into 1-hop ego networks.

6. Conclusion and Future Work

6.1 Key Findings

- **Exact Dataset Geometry Reconciled:** Successfully verified the SNAP baseline of 14,484 unique undirected edges and 5,241 connected vertices after eliminating 12 self-loops.
- **High Community Cohesion:** 79.34% of the entire network belongs to a single Giant Connected Component, demonstrating small-world academic community structure.
- **Degree vs. PageRank Discrepancy:** Local degree count is an insufficient proxy for scientific influence. Rank Delta analysis successfully uncovers high-prestige bridges connecting major subfields.
- **Scalable Decoupled Architecture:** Distributed batch processing in Apache Spark GraphX combined with sub-millisecond REST caching in FastAPI provides both computational scalability and interactive responsiveness.

6.2 Limitations and Future Work

Current limitations include binary unweighted edges and lack of temporal timestamping. Future enhancements include: (1) incorporating edge weights based on fractional paper author contributions, (2) tracking sliding-window temporal graph evolution across arXiv preprint years using Spark GraphFrames, and (3) applying NLP topic modeling on paper abstracts to classify components into specialized physical subdisciplines automatically.

7. References

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4. M. E. J. Newman. *The Structure of Scientific Collaboration Networks*. Proceedings of the National Academy of Sciences (PNAS), 98(2): 404–409, 2001.

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6. M. Franz et al. *Cytoscape.js: A Graph Theory Library for Visualisation and Analysis*. Bioinformatics, 32(2): 309–311, 2016.

7. OpenAlex Documentation. *OpenAlex Scholarly Metadata API Reference*. OurResearch, 2024. https://docs.openalex.org/

8. Appendix: Code and Reproducibility

8.1 Source Code Repository

All source code, Scala scripts, FastAPI backend modules, React frontend components, and dataset outputs are publicly accessible at:

GitHub Repository:
https://github.com/deepshika1211/-CollabSphere-Research-Collaboration-Intelligence-Platform.git

8.2 Instructions to Reproduce All Results

1. **Acquire Dataset:** `python scripts/download_dataset.py`
2. **Run Spark GraphX Pipeline:** `spark-shell --master "local[*]" --conf "spark.driver.memory=4g" -i scripts/run_graphx.scala`
3. **Execute Reference Graph Test:** `spark-shell --master "local[*]" -i scripts/verify_test_graph.scala`
4. **Execute Backend Test Suite:** `python backend/tests/run_tests.py (Passes 9/9 automated tests)`
5. **Launch Web Platform:** `Run .\run_all.bat (Starts Backend on :8000 and Frontend on :5173)`

8.3 Automated Verification Test Logs

Part 8 GraphX Reference Graph Test: Observed Vertices: 4 (Expected: 4) | Observed Edges: 4 (Expected: 4) | Degrees: d(1)=2, d(2)=2, d(3)=3, d(4)=1 [PASS] | Components: 1 Component [PASS] | PageRank: Node 3 highest (1.4668) [PASS].
FastAPI Integration Suite: Ran 9 tests in 0.209s — ALL 9 TESTS PASSED OK.